

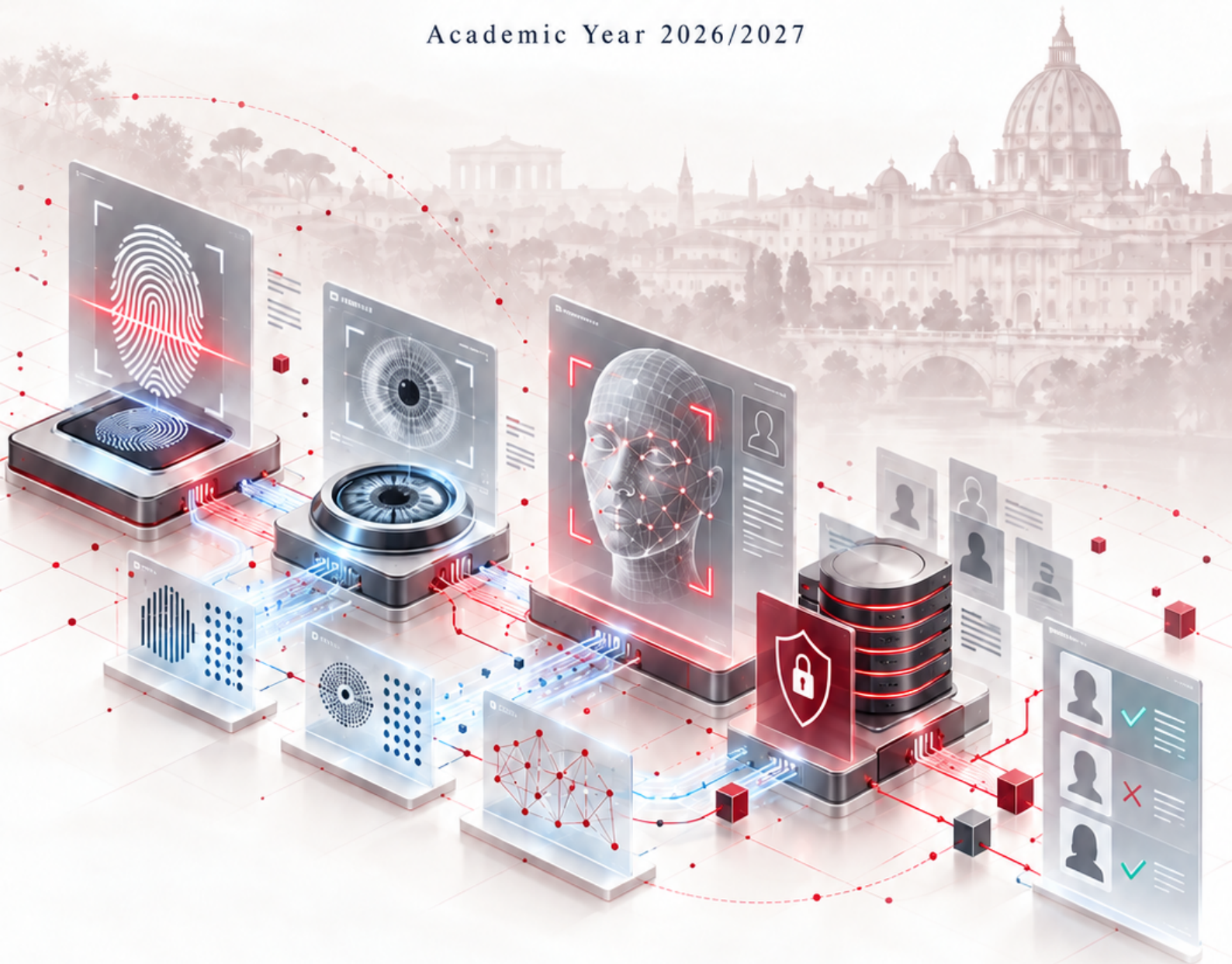
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Master's Degree in Computer Science

# BIOMETRIC SYSTEMS

## Course Notes

Academic Year 2026/2027



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Prepared by

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# 1 Introduction to Biometric Systems

Biometric recognition identifies or authenticates people from measurable physical or behavioural characteristics. The central engineering problem is to transform variable sensor observations into representations that remain discriminative across people and sufficiently stable for repeated use.

## 1.1 Biometrics as Pattern Recognition

A biometric system can be understood as a particular form of *pattern-recognition system*. A sample acquired from a person is represented through a collection of measurable characteristics, often summarized by a feature vector.

Two patterns are considered similar when an appropriate distance between their feature representations is sufficiently small, or equivalently when a similarity score is sufficiently large.

Three design questions determine whether a pattern-recognition pipeline can support reliable biometric matching:

1. Which distance or similarity measure is appropriate?
2. Which features are sufficiently informative and discriminative?
3. Which difference margin, or acceptance threshold, should be used to decide whether two patterns match?

Pattern recognition appears in different forms depending on the meaning of the classes.

In content-based image retrieval, classes can correspond to different object types, so the pattern representation separates, for example, a face from a flower.

In a more specific recognition problem, classes may correspond to subclasses of the same general type.

In biometrics, the classes are the *individuals themselves*: the pattern must distinguish one person from another.

### Key Concept

Biometrics uses pattern-recognition techniques in a setting where the identities of individual people are the classes to be discriminated.

## 1.2 What Is Biometrics?

The term *biometrics* derives from the Greek words *bios* (life) and *métron* (measure).

In a broad sense, it refers to the measurement and statistical analysis of biological data.

In technological applications, biometrics concerns the measurement and analysis of physical and/or behavioural characteristics for recognizing or authenticating a person.

### Definition – Biometrics

Biometrics is the automatic recognition of a person according to discriminative characteristics. In operational systems, measurable physical or behavioural traits are acquired, represented, compared, and used to support an identity decision.

The basic assumption is that each person is unique.

Turning this assumption into a working system requires three main steps:

- determine features that are sufficiently distinctive to identify a person;

- find reliable techniques for measuring those features;
- design reliable algorithms that can recognize or classify a person from the measured features.

### 1.3 Access Types

Biometric recognition can protect both physical and logical resources.

**Table 1.1:** Access types considered in biometric applications.

| Access type            | Examples                             |
|------------------------|--------------------------------------|
| <b>Physical access</b> | Room, building, controlled area.     |
| <b>Logical access</b>  | Electronic resources, critical data. |

### 1.4 Why Biometric Systems?

Traditional authentication commonly relies on two factors.

The first is *something one owns*, such as a card or document.

Such an object can be lost, stolen, or copied, and the system may effectively authenticate the object rather than the legitimate owner.

The second is *something one knows*, such as a password.

A password may be guessed, elicited from the user, or forgotten.

Moreover, a password that is easy to remember can also be easier to guess.

Biometric recognition adds a third possibility: *something one is*.

The identity evidence is derived from the person's own physical or behavioural characteristics.

**Table 1.2:** Authentication factors used in biometric authentication.

| Factor                     | Example         | Main limitation                                                                                                   |
|----------------------------|-----------------|-------------------------------------------------------------------------------------------------------------------|
| <b>Something one owns</b>  | Card, document  | Can be lost, stolen, or copied; the object rather than its owner may be authenticated.                            |
| <b>Something one knows</b> | Password        | Can be guessed, extracted from the user, or forgotten.                                                            |
| <b>Something one is</b>    | Biometric trait | Recognition is based on measurable characteristics of the person; accuracy and trait-specific limitations remain. |

### 1.5 Historical Background

Biometric recognition predates digital sensors and machine learning. Early identification systems already pursued the same two properties that remain central today: measurements should distinguish different people and should remain sufficiently stable over time.

#### 1.5.1 Bertillon and Bertillonage

In 1882, Alphonse Bertillon (1853–1914), chief of the identification service of the Paris police, introduced an anthropometric identification procedure later known as *Bertillonage*. The method recorded bodily measurements and descriptive characteristics such as hand shape, bust and limb measurements, head shape, facial details, and the corresponding identification card.

The method was not based on one measurement alone. Measurements were divided into discrete categories so that records could first be partitioned into smaller groups and only then compared

in detail. The historical scheme described 243 primary categories; further subdivision using seven eye categories together with hair colour produced 1,701 groupings. With a collection of 5,000 records, a primary category would contain roughly 20 cards, reducing a search from the whole archive to a small candidate set.

#### Example – The Will West case

In 1903, a newly admitted prisoner named Will West was measured using the Bertillon procedure. The resulting measurements led to an existing record for William West, whose measurements and appearance were strikingly similar. Further investigation established that William West was already incarcerated and that the two men were different individuals. The case illustrates the central limitation of coarse anthropometric categorization: two people can fall into nearly the same measurement classes even when their identities are different.

The rapid growth of early identification archives also made indexing an engineering concern. Anthropometry was introduced into a United States prison setting in 1887 by Major McClaughry. A National Bureau of Identification was established in 1896 and was described as a forerunner of later federal identification services. The historical account reports 16,000 Bertillon cards after one year, 24,000 after two years, and 131 prisoners recorded as first offenders who were found to have prior records. By 1900, the prison administration was formally permitted to accept Bertillon cards.

#### Remark – Historical context

Bertillonage is important as an early model of biometric identification, not as a modern biometric specification. Its lasting contribution is the attempt to formalize human identification through measurable, persistent, and discriminative characteristics.

### 1.5.2 Galton and Fingerprints

At the end of the nineteenth century, Francis Galton criticized anthropometric identification from a statistical point of view. In 1892 he introduced the notion of a fingerprint *minutia* and proposed an early fingerprint classification method.

In 1893, the UK Home Office recognized the individuality of fingerprints. Police organizations subsequently began acquiring and storing fingerprint records and developing systematic methods for visual comparison. Around 1900, the Galton–Henry classification system became an important basis for fingerprint-recognition practice in many police organizations.

## 1.6 Limitations and Integration with Traditional Systems

Biometric recognition does not always work perfectly.

Recognition failures motivate integration with traditional authentication factors.

Combining different forms of evidence can increase the security level compared with relying on only one factor.

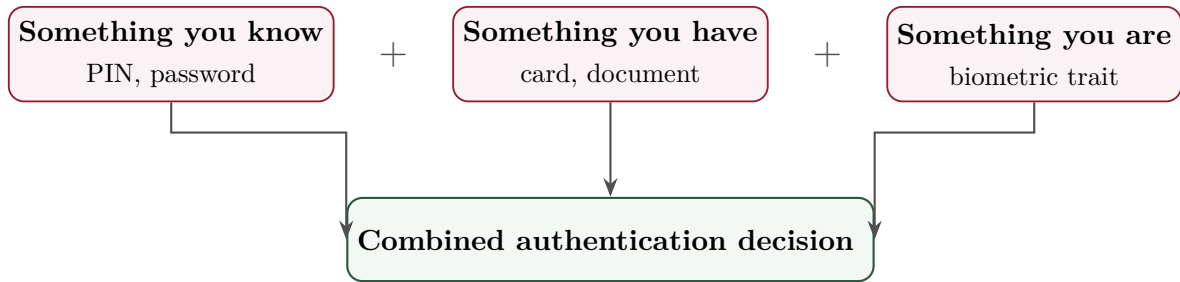
Figure 1.1 summarizes the three forms of authentication evidence discussed above.

## 1.7 Architecture of a Biometric System

A biometric system operates in two conceptually distinct phases: *enrollment* and *recognition*.

During enrollment, biometric data are captured and processed so that a reference template can be stored for future operations.

During recognition, newly acquired biometric data are converted into a probe template and compared with stored templates to support a decision.



**Figure 1.1:** Combination of knowledge, possession, and biometric evidence.

### Definition – Enrollment

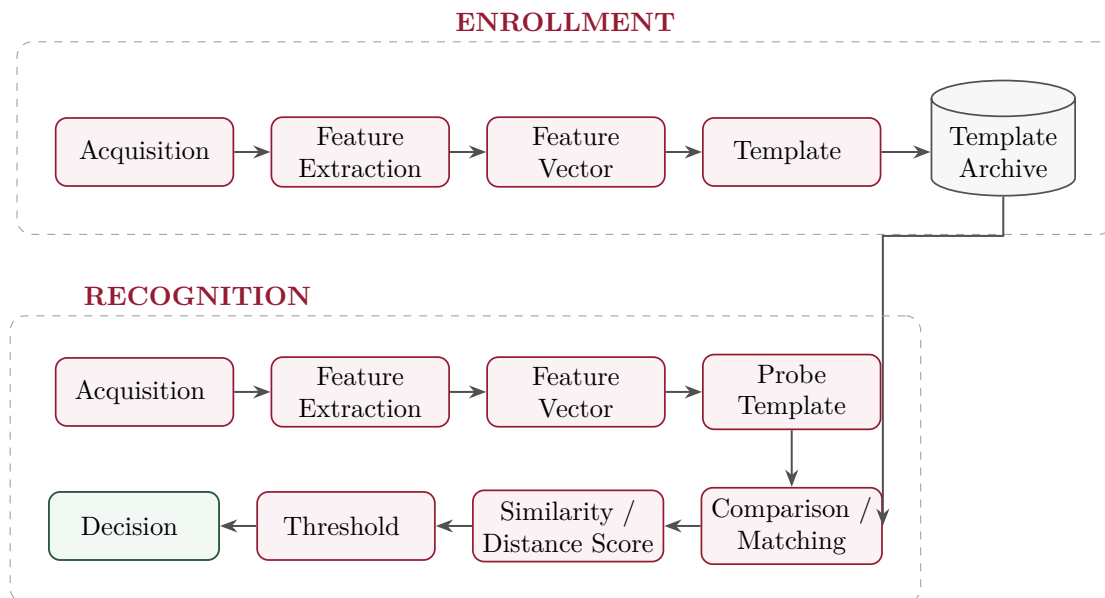
Enrollment is the capture and processing of a user’s biometric data for use by the system in subsequent authentication or recognition operations. The resulting reference templates populate the gallery.

### Definition – Recognition

Recognition is the capture and processing of biometric data in order to produce an authentication or identity decision from the result of matching a current template against stored template information.

### Definition – Probe and Gallery

A *probe* is a template submitted to the system for recognition. The *gallery* is the set of templates associated with enrolled subjects.



**Figure 1.2:** Conceptual architecture of enrollment and recognition in a biometric system.

Figure 1.2 separates the enrollment pipeline from the recognition pipeline and shows the stored template archive feeding the matcher.

Verification and identification use the same broad architecture but differ in the search space. Verification performs a 1:1 comparison with the template(s) associated with a claimed identity. Identification performs a 1: $N$  search over the gallery because no identity is claimed in advance.

## 1.8 Modules of a Biometric System

A biometric system can be organized into four functional modules.

**Table 1.3:** Main modules of a biometric system.

| Module                           | Role                                                                                                                               | Typical output                                                      |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| <b>Sensor module</b>             | Captures the biometric data from the subject.                                                                                      | Raw sample, such as an image, recording, or other sensed signal.    |
| <b>Feature extraction module</b> | Extracts the main characteristics from the acquired data. During enrollment it produces information used to form stored templates. | Feature vector or structured feature representation.                |
| <b>Matching module</b>           | Compares extracted features with stored templates.                                                                                 | One or more matching scores.                                        |
| <b>Decision module</b>           | Interprets matching results according to the operating rule.                                                                       | Acceptance, rejection, or an identity result depending on the task. |

## 1.9 Types of Users

Users can be classified along several dimensions.

**Table 1.4:** Operational categories of biometric-system users.

| Category                             | Meaning                                                                                                                             |
|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| <b>Cooperative / non-cooperative</b> | A cooperative user is interested in being recognized. A non-cooperative user may be indifferent or actively adverse to recognition. |
| <b>Public / private</b>              | Users may be customers/public users or employees/private users of the entity deploying the system.                                  |
| <b>Used / non-used</b>               | Describes how frequently the user interacts with the system: multiple times per day, daily, weekly, monthly, or only occasionally.  |
| <b>Aware / not aware</b>             | Describes whether the user is aware that a biometric recognition process is taking place.                                           |

## 1.10 Types of Acquisition Settings

In a *controlled* setting, capture conditions can be managed.

For face acquisition, for example, pose, illumination, and expression can be constrained.

Distortions can largely be avoided, defective templates can be rejected, and acquisition can be repeated.

In an *uncontrolled* or *undercontrolled* setting, capture conditions cannot be fully controlled.

Templates may exhibit different levels of distortion.

A defective template may still be detected and rejected, but the system may not have the opportunity to repeat the acquisition.

**Table 1.5:** Controlled and uncontrolled acquisition settings.

| Setting                               | Capture characteristics                                              | Consequence                                                                                   |
|---------------------------------------|----------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| <b>Controlled</b>                     | Pose, illumination, expression, and other conditions can be managed. | Distortion can often be reduced and acquisition may be repeated.                              |
| <b>Uncontrolled / undercontrolled</b> | Capture conditions cannot be fully managed.                          | Distortion may remain; defective templates can be detected but capture may not be repeatable. |

## 1.11 Types of Recognition Operations

### 1.11.1 Verification

In verification, the user claims an identity, possibly by presenting an ID card or another identifier.

The system performs a 1:1 matching operation against the reference information associated with that identity.

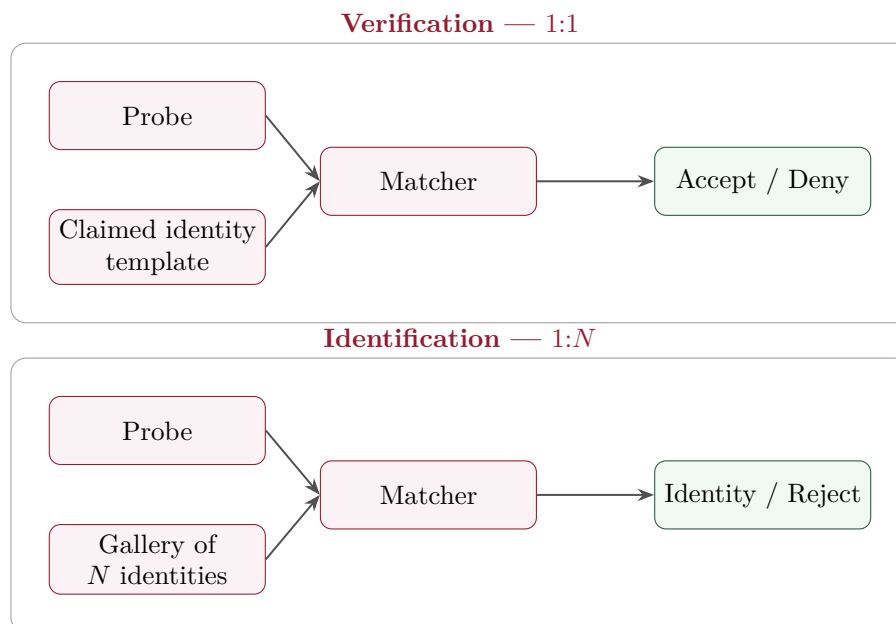
The output is typically *accept* or *deny*.

### 1.11.2 Identification

In identification, the user makes no identity claim.

The system must determine whether the probe corresponds to one of the subjects in the gallery, which requires a 1: $N$  operation.

The output is an identity when a suitable correspondence is found, with an explicit rejection option in open-set scenarios.



**Figure 1.3:** Verification compares a probe with a claimed identity, whereas identification searches a gallery.

The structural difference between the two operations is summarized in Figure 1.3.

### 1.11.3 Open-set, Closed-set, and Lists

In *open-set identification*, the system determines whether a probe  $p_i$  belongs to a subject represented in gallery  $G$ .

Some probes may not correspond to any enrolled subject, so the system needs a reject option. Errors include rejecting an enrolled subject, accepting a non-enrolled subject, or returning the wrong enrolled identity.

In *closed-set identification*, all probes belong to enrolled subjects.

The principal error is returning the wrong identity.

A *watch list* contains subjects for whom the system checks membership.

A *white list* contains subjects who are granted access, whereas a *black list* contains subjects who are rejected or may trigger an alarm.

## 1.12 Requirements for a Biometric Trait

**Table 1.6:** Requirements for a biometric trait.

| Requirement           | Meaning                                                                    |
|-----------------------|----------------------------------------------------------------------------|
| <b>Universality</b>   | The trait should be possessed by every person, except for rare exceptions. |
| <b>Uniqueness</b>     | Different people should be distinguishable according to the trait.         |
| <b>Permanence</b>     | The trait should remain sufficiently stable over time.                     |
| <b>Collectability</b> | The trait should be measurable by an appropriate sensor.                   |
| <b>Acceptability</b>  | People should not object to collection or measurement of the trait.        |

The X9.84-2003 standard context includes fingerprints, iris and retina, face, ear, hand or finger geometry, signature, keystroke behaviour, voice, and DNA or other biological traces.

These modalities may be associated with physiological, behavioural, or mixed features.

## 1.13 Strong and Soft Biometric Traits

Examples of *strong biometric traits* include iris, face, and fingerprint, whereas *soft biometric traits* include hair colour, facial shape, and gait.

Soft traits may lack uniqueness or long-term persistence; behavioural characteristics can also be affected by transient conditions such as mood or health. Even so, soft traits can reduce the search space by narrowing candidate identities.

## 1.14 Examples of Biometric Modalities

A biometric modality determines both what is sensed and how the resulting signal should be represented. Different modalities therefore require different acquisition procedures, features, and matching strategies.

### 1.14.1 Voice

Voice recognition can represent speech through acoustic feature vectors. A Gaussian Mixture Model (GMM) can model the distribution of those features: training observations are collected for several acoustic dimensions, their statistical distributions are estimated, and the resulting model is used to score a new utterance. A representative score is a log-likelihood of the form

$$\text{score} = \log p(\text{speech} \mid \text{speaker model}).$$

A larger likelihood indicates that the observed acoustic vectors are more compatible with the stored model.

### 1.14.2 Signature

A signature may be treated as a static image or as a dynamic behavioural signal. Dynamic acquisition can retain the trajectory of the pen together with time-varying measurements. Representative features include the  $x$ - and  $y$ -coordinates of the pen, pressure, pen azimuth in  $0^\circ$ – $359^\circ$ , and pen altitude in  $0^\circ$ – $90^\circ$ . The resulting template can therefore contain substantially more information than the final ink trace alone.

### 1.14.3 Fingerprint

Fingerprint recognition uses information at different spatial levels. First-level information captures global ridge-flow structure and broad fingerprint patterns. Second-level information focuses on local minutiae, such as ridge endings and bifurcations.

A typical processing chain begins from the fingerprint image, estimates the orientation field, identifies a region of interest, extracts ridges, thins the ridge representation, and finally extracts minutiae. Global ridge patterns include arches, tented arches, loops, and whorls. Matching then requires comparing the resulting local structures rather than merely comparing two raw images pixel by pixel.

### 1.14.4 Iris

Iris recognition first captures an eye image and localizes the iris region. One representative pipeline begins from a  $480 \times 640$  eye image. The annular iris texture is then normalized or *unwrapped*, encoded into a compact feature representation, and compared with enrolled templates. The comparison score is finally converted into an accept/reject decision. This pipeline separates geometric normalization from texture encoding so that matching can remain meaningful despite moderate changes in image scale or eye position.

### 1.14.5 Retina

Retinal recognition is based on the vascular pattern visible on the eyeground. The acquisition process captures the retina, localizes relevant vascular structures, derives a representation of the vessel pattern, and matches it against a stored template. The discriminative information is therefore associated with the geometry of the capillary network rather than with the external appearance of the eye.

### 1.14.6 Face

Face recognition admits several families of representations. Image-based approaches include methods based on Independent Component Analysis (ICA), neural networks, and eigenface-style representations. Feature-based approaches can compare structured facial landmarks, for example through elastic graph matching. Three-dimensional approaches can use 3D morphable models, while hybrid approaches combine multiple descriptions such as wavelet- or fractal-derived information.

**Table 1.7:** Representative biometric modalities and the information used for matching.

| Modality    | Representative information                                          | Typical comparison idea                                   |
|-------------|---------------------------------------------------------------------|-----------------------------------------------------------|
| Voice       | Acoustic feature vectors and their statistical distribution         | Likelihood or vector-based comparison.                    |
| Signature   | Static shape or dynamic pen trajectory, pressure, azimuth, altitude | Image, sequence, or feature-vector matching.              |
| Fingerprint | Global ridge flow and local minutiae                                | Structured correspondence between ridge/minutia features. |
| Iris        | Normalized iris texture encoding                                    | Template comparison after localization and normalization. |
| Retina      | Vascular pattern on the eyeground                                   | Comparison of extracted vessel structures.                |
| Face        | Appearance, landmarks, learned features, or 3D structure            | Image-, feature-, embedding-, or model-based comparison.  |

### 1.15 Benefits and Drawbacks of Biometric Traits

Biometric traits provide a natural form of authentication because the user need only appear in person.

They cannot be forgotten or lent in the same way as a password or token.

These advantages are accompanied by several drawbacks:

- biometric systems do not provide 100% accuracy;
- some users cannot be reliably recognized by some technologies, for example when fingerprints are heavily worn or damaged;
- traits may change over time;
- biometric devices may be unreliable under some circumstances;
- a copied biometric trait cannot simply be replaced in the same way as a compromised password.

**Remark**

A biometric characteristic is physically bound to the person, but a digital or physical representation of that characteristic can still be copied or compromised. Unlike a password, the underlying trait cannot simply be replaced after compromise.

### 1.16 Chapter Summary

Biometric systems treat personal identity as a pattern-recognition problem. A complete system must acquire a suitable trait, extract a stable representation, compare that representation with enrolled templates, and convert the resulting score into a decision. Verification performs a 1:1 comparison for a claimed identity, whereas identification searches a 1:*N* gallery and may additionally require a reject option. The practical quality of a modality depends on universality, uniqueness, permanence, collectability, and acceptability, while real deployments must also account for acquisition conditions, user cooperation, imperfect recognition, and the consequences of biometric compromise.

## 2 Performance Evaluation of Biometric Systems

A biometric matcher does not return certainty; it returns evidence, usually as a similarity or distance score. Performance evaluation studies how those scores behave for genuine and impostor comparisons and how an operating threshold converts them into correct decisions and errors.

### 2.1 Sources of Recognition Difficulty

Biometric recognition is difficult because the same person does not always produce identical samples, while different people can sometimes appear very similar according to the measured trait.

- **Large intra-class variation:** samples from the same person may vary substantially, for example because of pose or facial expression.
- **Small inter-class variation:** samples from different people may be very similar, as illustrated by twins or close relatives.
- **Noisy or distorted acquisitions:** poor fingerprint quality or non-uniform illumination can degrade the sample.
- **Non-universality:** some people may not provide a usable sample for a particular modality. About 4% of the population is reported to present poor-quality fingerprints, with the issue being more frequent in some groups such as elderly people.
- **Spoofing and attacks:** attacks may target different points of the biometric processing pipeline.

Potential attack points span the complete biometric pipeline: the sensor, feature extractor, communication channels, matcher, stored templates, and final decision path.

Examples include presenting a fake biometric, replaying old data, overriding feature extraction, synthesizing feature vectors, modifying stored templates, overriding the matcher, intercepting channels, or overriding the final decision.

#### Key Concept

Performance is not determined only by the matcher.  
It depends on acquisition quality, representation quality, population variability, the operating threshold, and the integrity of the complete biometric pipeline.

### 2.2 Samples, Hand-crafted Features, and Templates

#### Definition – Sample

A sample is the raw captured biometric data, such as an image, a voice recording, or a fingerprint acquisition.

#### Definition – Hand-crafted Features

Hand-crafted features are characteristics manually designed by a data scientist and extracted from raw samples.

**Definition – Template**

A template is a collection of features extracted from the raw biometric data and used for comparison.

Templates may have different mathematical structures.

Representative template structures include:

- a histogram of relevant image values such as grey levels;
- a vector of measurements, such as a collection of anthropometric measurements;
- a time series, such as acceleration values recorded along the three accelerometer axes;
- a set of fingerprint minutiae represented by triplets

$$\{(x_1, y_1, \theta_1), \dots, (x_n, y_n, \theta_n)\},$$

where the coordinates identify relevant points and  $\theta_i$  represents the local ridge-tangent direction.

**2.3 Comparing Templates**

The comparison procedure depends on the representation.

Vector templates can often be compared with standard distances or similarities, whereas structured representations may require specialized matching.

**2.3.1 Euclidean Distance**

For vectors  $\mathbf{p}, \mathbf{q} \in \mathbb{R}^n$ , the Euclidean distance is

$$d(\mathbf{p}, \mathbf{q}) = \|\mathbf{p} - \mathbf{q}\|_2 = \sqrt{\sum_{i=1}^n (p_i - q_i)^2}. \quad (2.1)$$

Here  $n$  is the feature-space dimension,  $p_i$  and  $q_i$  are corresponding vector components, and  $\|\cdot\|_2$  denotes the Euclidean norm. A smaller Euclidean distance indicates greater similarity.

**2.3.2 Cosine Similarity**

The cosine similarity between vectors  $\mathbf{A}$  and  $\mathbf{B}$  is

$$S_C(\mathbf{A}, \mathbf{B}) = \cos(\theta) = \frac{\mathbf{A}^\top \mathbf{B}}{\|\mathbf{A}\|_2 \|\mathbf{B}\|_2} = \frac{\sum_{i=1}^n A_i B_i}{\sqrt{\sum_{i=1}^n A_i^2} \sqrt{\sum_{i=1}^n B_i^2}}. \quad (2.2)$$

Here  $\theta$  is the angle between the two vectors,  $\mathbf{A}^\top \mathbf{B}$  is their dot product, and  $\|\mathbf{A}\|_2, \|\mathbf{B}\|_2$  are their Euclidean norms. Unlike a distance, larger cosine similarity values represent greater similarity.

**2.3.3 Pearson Correlation**

Pearson correlation can be used as a similarity measure for histograms or sets of paired values.

Given paired samples  $(x_i, y_i)$ ,  $i = 1, \dots, n$ ,

$$r_{xy} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}}. \quad (2.3)$$

An algebraically equivalent form is

$$r_{xy} = \frac{n \sum_i x_i y_i - (\sum_i x_i)(\sum_i y_i)}{\sqrt{n \sum_i x_i^2 - (\sum_i x_i)^2} \sqrt{n \sum_i y_i^2 - (\sum_i y_i)^2}}. \quad (2.4)$$

Here  $\bar{x}$  and  $\bar{y}$  are the sample means and  $n$  is the number of paired observations. The coefficient  $r_{xy}$  lies in the usual correlation range and is interpreted here as a similarity measure.

### 2.3.4 Bhattacharyya Distance

For discrete probability distributions  $P$  and  $Q$  over the same domain  $\mathcal{X}$ , the Bhattacharyya coefficient is

$$BC(P, Q) = \sum_{x \in \mathcal{X}} \sqrt{P(x)Q(x)}. \quad (2.5)$$

The corresponding distance is

$$D_B(P, Q) = -\ln(BC(P, Q)). \quad (2.6)$$

Here  $P(x)$  and  $Q(x)$  are the probability masses assigned to  $x \in \mathcal{X}$ ,  $BC(P, Q)$  is the Bhattacharyya coefficient, and  $D_B(P, Q)$  is the corresponding distance. This measure is useful for comparing histogram-like or probabilistic representations.

### 2.3.5 Dynamic Time Warping

Dynamic Time Warping (DTW) is presented as a distance method for time series.

The motivating example compares two repetitions of a walking sequence recorded with motion capture.

Even when walking speeds differ, spatial trajectories may remain highly similar.

DTW allows a non-linear alignment of time indices before comparison.

### 2.3.6 Learned Features and Embeddings

In deep-learning systems, the final classification layer, typically a softmax layer, can be omitted when the goal is to extract the internal *embedding* that would otherwise feed the classifier.

These embeddings can then be treated as vectors and compared using vector-based similarity or distance measures.

Some representations require more than direct vector comparison.

Fingerprint minutiae, for example, require finding a suitable pairing between points before the match can be evaluated.

## 2.4 Similarity, Distance, and Acceptance Thresholds

Once a similarity or distance value has been obtained, it is compared with an acceptance threshold in verification and open-set identification.

For a similarity score  $s$  and threshold  $t$ ,

$$\text{accept} \iff s \geq t. \quad (2.7)$$

For a distance  $d$ ,

$$\text{accept} \iff d \leq t. \quad (2.8)$$

Reasoning with similarities and distances is symmetric after reversing the inequality and the interpretation of larger and smaller values.

**Table 2.1:** Comparison methods for representative biometric template structures.

| Method                        | Type          | Typical use                                                          |
|-------------------------------|---------------|----------------------------------------------------------------------|
| <b>Euclidean distance</b>     | Distance      | Vector templates.                                                    |
| <b>Cosine similarity</b>      | Similarity    | Vector templates and embeddings.                                     |
| <b>Pearson correlation</b>    | Similarity    | Histograms or paired point sets.                                     |
| <b>Bhattacharyya distance</b> | Distance      | Histogram or distribution comparison.                                |
| <b>Dynamic Time Warping</b>   | Distance      | Time-series comparison with temporal misalignment.                   |
| <b>Structured matching</b>    | Task-specific | Feature sets such as fingerprint minutiae where pairing is required. |

## 2.5 Verification Outcomes

A verification claim is accepted when the best matching score for the claimed identity satisfies the threshold rule.

Four cases arise.

**Table 2.2:** Verification outcomes and common biometric terminology.

| True situation | System decision | Biometric term          | Alternative term        |
|----------------|-----------------|-------------------------|-------------------------|
| Genuine claim  | Accept          | Genuine Acceptance (GA) | Genuine Match (GM)      |
| Genuine claim  | Reject          | False Rejection (FR)    | False Non-Match (FNM)   |
| Impostor claim | Reject          | Genuine Rejection (GR)  | Genuine Non-Match (GNM) |
| Impostor claim | Accept          | False Acceptance (FA)   | False Match (FM)        |

### Remark – Terminology note

Type I and Type II depend on the choice of the null hypothesis. With  $H_0 = \text{different person}$ , as defined below, false acceptance is a Type I error and false rejection is a Type II error. The opposite labels require  $H_0 = \text{same person}$ .

The biometric names FAR and FRR are the less ambiguous terms to use when interpreting the hypothesis notation below.

### Remark – Uncertain cases

A practical possibility between automatic acceptance and rejection is to route an uncertain score region to a human operator. Such a three-way policy is application-specific and does not change the definitions of GA, GR, FA, and FR used for binary evaluation.

## 2.6 False Acceptance Rate and False Rejection Rate

### Performance Metric – False Acceptance Rate (FAR/FMR)

The False Acceptance Rate (FAR), also called False Match Rate (FMR), is the fraction or probability of impostor recognition operations in which the impostor is incorrectly accepted.

For example,

$$\text{FAR} = 0.1\%$$

means that, on average, one out of every 1000 impostor attempts is successful.

### Performance Metric – False Rejection Rate (FRR/FNMR)

The False Rejection Rate (FRR), also called False Non-Match Rate (FNMR), is the fraction or probability of genuine recognition operations in which the genuine user is incorrectly rejected.

For example,

$$\text{FRR} = 0.05\%$$

means that, on average, one out of every 2000 authorized access attempts is incorrectly rejected.

## 2.7 Ground Truth and Evaluation Notation

Evaluation experiments use labelled samples, so the true identity is known through a ground-truth function  $\text{id}(\cdot)$ .

| Notation                  | Meaning                                                                                           |
|---------------------------|---------------------------------------------------------------------------------------------------|
| $\text{id}(p_j)$          | True identity of probe $p_j$ .                                                                    |
| $\text{id}(g_x)$          | Identity associated with gallery template $g_x$ .                                                 |
| $\text{topMatch}(p_j, i)$ | Best match between probe $p_j$ and the gallery template(s) associated with claimed identity $i$ . |
| $s(t_1, t_2)$             | Matching score between two templates.                                                             |

Let  $G = \{g_i\}$  denote the gallery templates. Let  $P_G$  be the set of probes belonging to subjects represented in the gallery and  $P_N$  the set of probes belonging to subjects not represented in the gallery.

For a genuine claim,

$$g_x = \text{topMatch}(p_j, \text{id}(p_j)), \quad s_{xj} = s(g_x, p_j), \quad p_j \in P_G. \quad (2.9)$$

For an impostor claim,

$$g_x = \text{topMatch}(p_j, i), \quad i \neq \text{id}(p_j), \quad s_{xj} = s(g_x, p_j). \quad (2.10)$$

Using the similarity-threshold convention,

$$\text{FRR}(t) = \frac{|\{p_j : s_{xj} < t, \text{id}(g_x) = \text{id}(p_j)\}|}{|\{p_j : \text{id}(g_x) = \text{id}(p_j)\}|}. \quad (2.11)$$

Similarly,

$$\text{FAR}(t) = \frac{|\{p_j : s_{xj} \geq t, \text{id}(g_x) \neq \text{id}(p_j)\}|}{|\{p_j : \text{id}(g_x) \neq \text{id}(p_j)\}|}. \quad (2.12)$$

Two impostor scenarios are possible. An impostor may itself be registered but claim another identity, or may be completely outside the gallery.

For verification error computation, both are false claims.

## 2.8 Genuine and Impostor Score Distributions

A score is *genuine* when it results from matching two samples from the same enrolled individual.

A score is *impostor* when it results from matching samples associated with different identities.

The decision problem can be written through the hypotheses

$H_0$  : different person,

$H_1$  : same person.

Possible decisions are

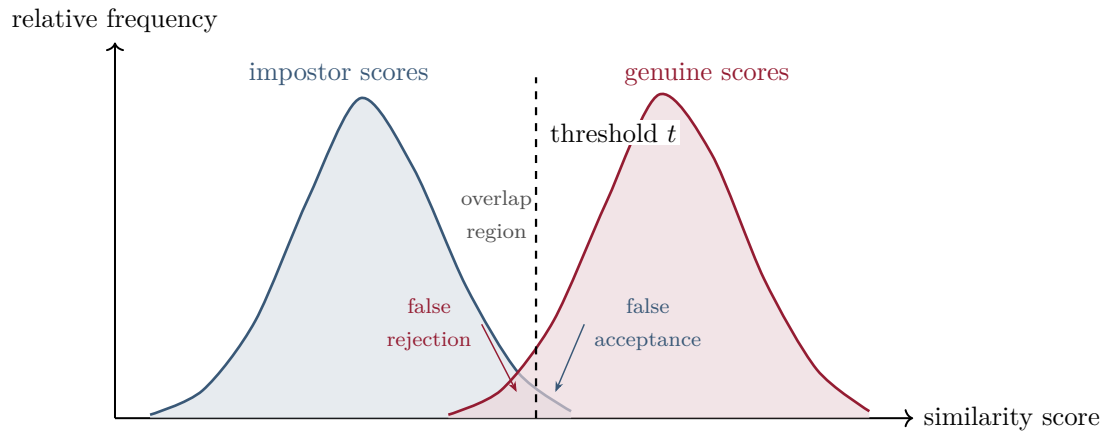
$D_0$  : decide different person,

$D_1$  : decide same person.

The corresponding probabilities are

$$\text{FAR} = \Pr(D_1 \mid H_0), \quad \text{FRR} = \Pr(D_0 \mid H_1). \quad (2.13)$$

Figure 2.1 gives a schematic view of the two score distributions and the effect of a similarity threshold.



**Figure 2.1:** Schematic genuine and impostor similarity-score distributions. Numerical coordinates are used only to draw the conceptual shapes; they do not represent experimental measurements.

## 2.9 GAR, EER, ZeroFAR, and ZeroFRR

The Genuine Acceptance Rate (GAR), also called Genuine Match Rate (GMR), is

$$\text{GAR} = 1 - \text{FRR}. \quad (2.14)$$

The Equal Error Rate (EER) is the error level obtained at a threshold  $t^*$  where

$$\text{FAR}(t^*) = \text{FRR}(t^*). \quad (2.15)$$

Two limiting operating points are also useful:

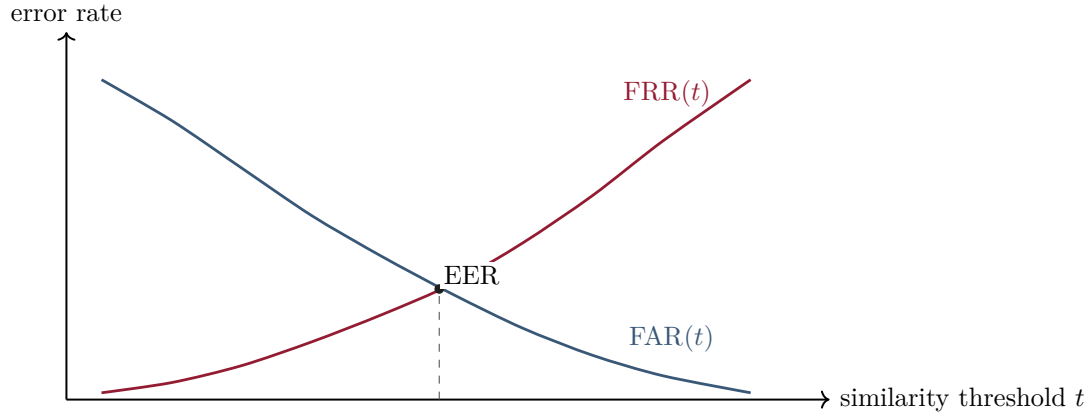
- **ZeroFRR / ZeroFNMR:** the FAR obtained when FRR is zero;
- **ZeroFAR / ZeroFMR:** the FRR obtained when FAR is zero.

For distance-based matching, an excessively low threshold produces many genuine rejections, while an excessively high threshold produces many false acceptances.

Under the similarity-threshold convention used in Equations (2.7), (2.11), and (2.12), the direction is reversed: increasing the threshold makes acceptance stricter, so FAR decreases while FRR increases.

The EER threshold is a convenient and common comparison point because it balances the two verification error rates. It is not automatically the best operating threshold for a deployed application: the operating threshold must follow the relative cost of false acceptances and false rejections.

Figure 2.2 shows the corresponding threshold trade-off using the same similarity-threshold convention as Equations (2.7), (2.11), and (2.12).



**Figure 2.2:** Schematic FAR/FRR trade-off under a *similarity-threshold* convention: increasing  $t$  makes acceptance stricter, so FAR decreases while FRR increases. The intersection illustrates EER; no experimental values are implied.

## 2.10 ROC and DET Curves

### Performance Metric – ROC

The Receiver Operating Characteristic (ROC) plots Genuine Acceptance Rate

$$\text{GAR} = 1 - \text{FRR}$$

against False Acceptance Rate (FAR) as the threshold varies.

### Performance Metric – DET

The Detection Error Trade-off (DET) curve plots False Rejection Rate (FRR) against False Acceptance Rate (FAR).

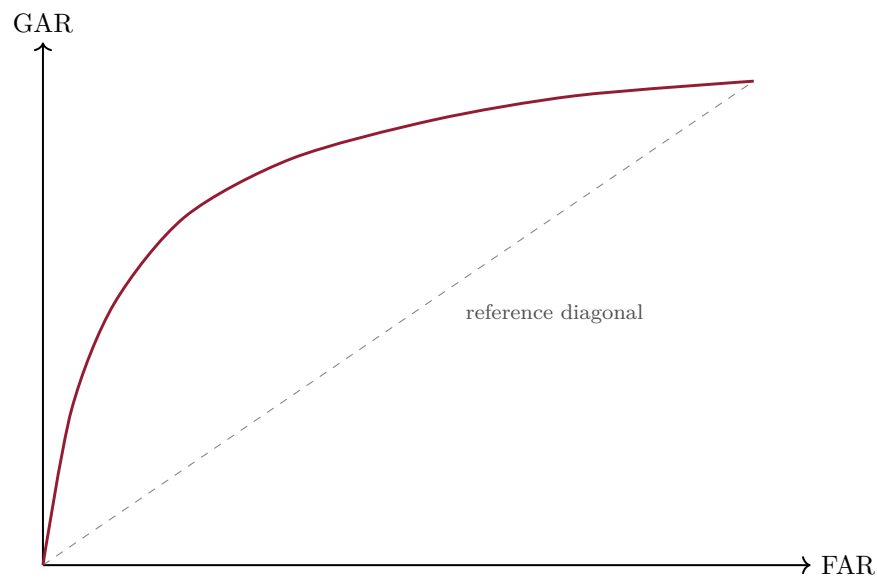
The standard DET uses normal-deviate (probit) axes: the plotted coordinates are  $\Phi^{-1}(\text{FAR})$  and  $\Phi^{-1}(\text{FRR})$ , where  $\Phi$  is the standard normal CDF. This is not a logarithmic scale.

Each operating threshold corresponds to a point on the relevant performance curve.

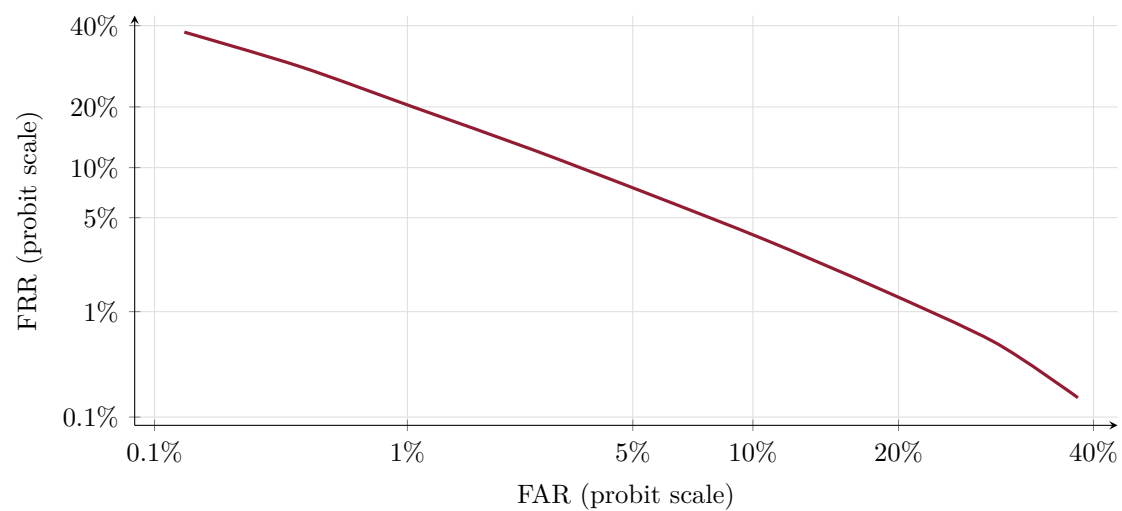
The two curve types are illustrated schematically in Figures 2.3 and 2.4.

### Key Concept

For ROC curves, better systems tend toward high GAR at low FAR, i.e. toward the upper-left operating region. For DET curves, better systems tend toward simultaneously low FAR and low FRR, i.e. toward the lower-left region. Curve comparisons must still be made at operating points relevant to the intended application.



**Figure 2.3:** Schematic ROC representation with FAR on the horizontal axis and GAR on the vertical axis. The curve is illustrative only and contains no measured performance values.



**Figure 2.4:** Schematic DET representation. Both axes are normal-deviate (probit) transforms of the corresponding error probabilities; the tick labels show the original FAR and FRR percentages. The curve is conceptual and does not encode experimental measurements.

## 2.11 Open-set Identification Performance

In open-set identification, such as a watch-list application, the system addresses two questions:

1. Is the probe subject represented in the gallery?
2. If so, who is the probe subject?

Open-set identification combines ranked similarity scores with an acceptance threshold.

For a probe belonging to a gallery subject, correct operation requires both detection above the threshold and correct ranking of the true identity.

For a probe outside the gallery, correct operation requires that no gallery candidate incorrectly triggers an alarm.

Typical open-set applications include comparing people observed in an airport against a watch list or comparing an unidentified person against a missing-person database. These applications require both a membership decision and, when a match is detected, an identity decision.

### Example – Thresholded ranked scores

Suppose four candidates obtain scores

$$0.90, 0.86, 0.60, 0.40.$$

With a similarity threshold of 0.85, two candidates exceed the threshold.

If the top-ranked candidate is the correct identity, the operation is both a correct detection and a correct identification.

Raising the threshold to 0.95 produces no detection.

If the correct identity is not the highest-scoring candidate, detection can occur without correct identification. For example, with scores 0.86, 0.80, 0.60, 0.40 and threshold 0.75, detection occurs, but the operation is not a correct detect-and-identify if the true identity is the second-ranked candidate.

For a probe from outside the gallery, scores 0.80, 0.70, 0.60, 0.40 with threshold 0.85 produce no alarm and therefore a correct rejection. Lowering the threshold to 0.75 makes the highest score trigger an alarm, which is a false alarm because the probe has no mated gallery identity.

For probes from  $P_G$ , repeated trials estimate how often the system correctly detects and identifies the subject.

This quantity is the *correct detect and identify rate*.

For probes from  $P_N$ , repeated trials estimate how often an incorrect alarm occurs.

This is the *false alarm rate*.

### 2.11.1 Rank and DIR

Let  $\text{rank}(p_j)$  denote the position in the ranked list at which the first template for the correct identity appears.

The Detection and Identification Rate,  $\text{DIR}(t, k)$ , expresses the probability that a gallery probe is detected above operating threshold  $t$  and that the correct identity appears within the first  $k$  ranked candidates. For similarity scores, a precise form is

$$\text{DIR}(t, k) = \frac{\left| \left\{ p_j \in P_G : \text{rank}(p_j) \leq k, \max_{g_i \in G: \text{id}(g_i) = \text{id}(p_j)} s(g_i, p_j) \geq t \right\} \right|}{|P_G|}. \quad (2.16)$$

Here  $k$  is the maximum accepted rank,  $t$  is the similarity threshold, and the inner maximum selects the best mated gallery template for probe  $p_j$ .

At rank 1, the corresponding False Negative Identification Rate (FNIR) is

$$\text{FNIR}(t) = 1 - \text{DIR}(t, 1). \quad (2.17)$$

The explicit threshold dependence is important in open-set evaluation because changing  $t$  changes both the detection probability and the false-alarm rate.

The false-alarm counterpart is the False Positive Identification Rate (FPIR), also described as FAR or false alarm rate in a watch-list context. For non-enrolled probes,

$$\text{FPIR}(t) = \frac{|\{p_j \in P_N : \max_{g_i \in G} s(g_i, p_j) \geq t\}|}{|P_N|}. \quad (2.18)$$

Thus  $G$  is the gallery,  $P_N$  is the set of non-gallery probes, and an alarm occurs whenever at least one gallery score reaches the similarity threshold. The same quantity is also described using FAR/false-alarm terminology in some contexts; FPIR is the more task-specific name in open-set identification.

At rank 1, an open-set equal-error operating point can be described by  $\text{FNIR}(t) = \text{FPIR}(t)$ , i.e. by equalizing the paired open-set error probabilities.

### 2.11.2 Open-set ROC and Operating Regions

The detect-and-identify rate and false-alarm rate vary in opposite directions with the threshold.

Raising a similarity threshold tends to reduce both successful detection and false alarms.

Lowering the threshold tends to increase both.

Five practical operating preferences can be distinguished:

1. extremely low false alarm;
2. extremely high detection and identification probability;
3. low false alarm with lower detection and identification;
4. high detection and identification with higher false alarms;
5. no fixed threshold, with ranked results returned for investigation.

## 2.12 Closed-set Identification Performance

Closed-set identification is a special case in which every probe is known to have a corresponding identity in the gallery.

The membership question is therefore already answered.

Pure closed-set operation is relatively restrictive in practice. A system may be described broadly as an identification system while operationally behaving like a watch list; the FBI Integrated Automated Fingerprint Identification System (IAFIS) is a representative example of this distinction.

The remaining issue is how highly the correct identity is ranked.

If the correct identity has the highest similarity score, the system has achieved correct identification at rank 1.

If the correct identity has the second-highest score, it is within the top two results and contributes to correct identification at rank 2.

**Example – Closed-set ranks**

With similarity scores 0.90, 0.86, 0.60, 0.40, a correct identity in the first position gives a rank-1 identification. With scores 0.86, 0.80, 0.60, 0.40, if the correct identity is the second-ranked candidate, the result contributes to identification at rank 2 even though it is not correct at rank 1. More generally, rank  $k$  asks whether the correct identity appears anywhere within the first  $k$  candidates.

**Performance Metric – Cumulative Match Score (CMS)**

The CMS at rank  $k$  is the probability that the correct identity is found among the first  $k$  positions of the ranked candidate list.

Equivalently, it can be estimated as the fraction of test probes whose correct identity appears within the top  $k$  results.

**Performance Metric – Cumulative Match Characteristic (CMC)**

A CMC curve plots the cumulative match score as a function of rank.

CMS at rank 1 gives the probability that the correct identity is the first returned candidate.

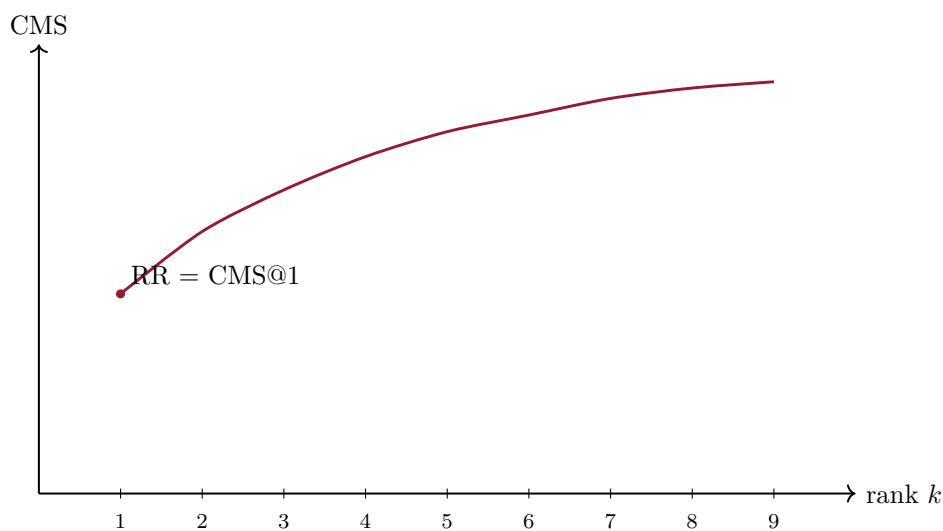
CMS at rank 2 gives the probability that it appears in either of the first two positions.

More generally, CMS at rank  $k$  gives the probability that the correct identity appears among the first  $k$  positions.

When the rank reaches the gallery size, the cumulative probability reaches 1 because the correct identity must occur somewhere in the complete closed-set list.

CMS at rank 1 is also called the *Recognition Rate (RR)*.

Figure 2.5 visualizes this cumulative ranking behaviour.



**Figure 2.5:** Schematic cumulative match characteristic (CMC) for closed-set identification. The curve illustrates cumulative ranking behaviour only; no dataset-specific values are implied.

## 2.13 Avoiding Confusion with Machine-Learning Metrics

Biometric error measures should not be confused with precision, recall, and F-score.

In standard machine-learning notation, let  $TP$ ,  $FP$ ,  $TN$ , and  $FN$  denote true positives, false positives, true negatives, and false negatives, respectively. Then

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}. \quad (2.19)$$

The F-score combines precision and recall; it is therefore also conceptually different from FAR and FRR.

In biometric acceptance/rejection terms, precision can be compared with

$$\frac{GA}{FA + GA},$$

while recall can be compared with

$$\frac{GA}{GA + FR}.$$

The biometric error rates of primary interest are instead

$$\text{FRR} = \frac{FR}{GA + FR}, \quad \text{FAR} = \frac{FA}{FA + GR}. \quad (2.20)$$

Under the mapping acceptance  $\leftrightarrow$  positive and rejection  $\leftrightarrow$  negative,

$$\text{FRR} = \frac{FN}{TP + FN}, \quad \text{FAR} = \frac{FP}{FP + TN}. \quad (2.21)$$

Thus FRR corresponds to the false-negative rate or miss rate, while FAR corresponds to the false-positive rate or fall-out.

Precision and recall therefore do not express the same error statistics as FAR and FRR.

## 2.14 Identification versus Re-identification

Biometric identification and re-identification (re-ID) address related but different tasks.

**Table 2.3:** Identification and re-identification as related recognition tasks.

| Task                     | Objective                                                                                                                                                                                    | Time perspective / context                                 |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| <b>Identification</b>    | Determine the identity of an unknown subject.                                                                                                                                                | Medium-to-long-term identity recognition.                  |
| <b>Re-identification</b> | Match the presence of the same person across different cameras, locations, video segments, image sequences, or event sets using appearance, body shape, behaviour, and tracking information. | Short-term association across non-overlapping time slices. |

Re-identification is evaluated with metrics adapted to ranked retrieval.

CMC is appropriate when there is a single mated gallery image for each query.

When several mated images may exist, Average Precision (AP) and mean Average Precision (mAP) are used.

AP is described as the area under a precision–recall curve computed over thresholds, while mAP averages AP over queries.

Re-identification can connect back to identification when a person is repeatedly found in video and the resulting identity evidence is then checked against a watch list.

## 2.15 Chapter Summary

Performance evaluation connects matcher scores to operational error probabilities. The principal measures for verification and identification are consolidated below.

**Table 2.4:** Compact summary of central biometric performance measures.

| Measure                   | Main task                 | Meaning                                                                              |
|---------------------------|---------------------------|--------------------------------------------------------------------------------------|
| <b>FAR / FMR</b>          | Verification              | Probability or fraction of impostor claims incorrectly accepted.                     |
| <b>FRR / FNMR</b>         | Verification              | Probability or fraction of genuine claims incorrectly rejected.                      |
| <b>GAR / GMR</b>          | Verification              | Genuine acceptance probability; $1 - \text{FRR}$ .                                   |
| <b>EER</b>                | Verification / open set   | Operating point where paired error rates are equal.                                  |
| <b>DIR</b>                | Open-set identification   | Detection and identification performance at a chosen rank.                           |
| <b>FNIR</b>               | Open-set identification   | False negative identification rate.                                                  |
| <b>FPIR</b>               | Open-set identification   | False positive identification or false alarm probability.                            |
| <b>CMS@<math>k</math></b> | Closed-set identification | Probability that the correct identity appears among the first $k$ ranked candidates. |
| <b>CMC</b>                | Closed-set identification | Curve of CMS values across ranks.                                                    |
| <b>RR</b>                 | Closed-set identification | Recognition rate, equal to CMS at rank 1.                                            |

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# Quick Review

The most exam-relevant relationships are grouped below for rapid review.

## Key Concept

### Foundations and architecture

- Biometrics is pattern recognition in which the classes are individual people.
- A biometric system uses acquisition, feature extraction, matching, and decision modules.
- Enrollment creates reference templates; recognition compares a probe with stored templates.
- Verification is 1:1; identification is 1:N.
- A useful trait should provide universality, uniqueness, permanence, collectability, and acceptability.
- Controlled acquisition reduces nuisance variation; uncontrolled acquisition may not permit recapture.

## Remark – Matching rules

similarity: accept  $\iff s \geq t$ ,

distance: accept  $\iff d \leq t$ .

Increasing a *similarity* threshold makes acceptance stricter; increasing a *distance* threshold makes acceptance looser.

## Performance Metric – Verification

- FAR/FMR: impostor claim incorrectly accepted.
- FRR/FNMR: genuine claim incorrectly rejected.
- GAR/GMR:  $1 - \text{FRR}$ .
- EER: operating point where FAR and FRR are equal.
- Genuine and impostor score distributions overlap; the threshold divides that overlap into false acceptances and false rejections.
- ROC plots GAR versus FAR. DET plots FRR versus FAR on normal-deviate (probit) axes.

## Performance Metric – Identification

- Open set asks both whether the probe belongs to the gallery and, if so, who it is.
- $\text{DIR}(t, k)$ : successful detection with the correct identity within the first  $k$  ranks.
- $\text{FNIR}(t) = 1 - \text{DIR}(t, 1)$ .
- FPIR: false alarm probability for non-gallery probes.
- Closed set assumes every probe has a gallery mate and is evaluated by rank.
- $\text{CMS}@k$ : probability that the correct identity is within the first  $k$  candidates.
- CMC plots CMS versus rank;  $\text{CMS}@1$  is the Recognition Rate.

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                     |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Remark – Do not confuse                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                     |
| <ul style="list-style-type: none"><li>• FAR and FRR are not precision and recall. FAR corresponds to FPR/fall-out; FRR corresponds to FNR/miss rate.</li><li>• Type I/II labels depend on the null hypothesis. FAR/FRR terminology is safer.</li><li>• Standard DET axes are probit, not ordinary logarithmic axes.</li><li>• EER is a comparison point, not a universally optimal deployment threshold.</li><li>• A raw sample is distinct from the feature representation or template extracted from it.</li></ul>                                                                                                                                      |                                                                                                                                     |
| Key Concept                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                     |
| Modalities and representations                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                     |
| <ul style="list-style-type: none"><li>• Strong biometric traits aim at identity discrimination; soft traits mainly narrow the candidate set.</li><li>• Vector templates may use Euclidean distance or cosine similarity; histograms/distributions may use correlation or Bhattacharyya distance; time series may use DTW.</li><li>• Learned embeddings can be compared as vectors; structured templates such as fingerprint minutiae require task-specific correspondence.</li><li>• Re-identification is ranked retrieval across observations: CMC is useful for a single mated gallery image, while AP/mAP address multiple relevant matches.</li></ul> |                                                                                                                                     |
| Remark – Compact equation sheet                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                     |
| Verification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | $\text{FAR} = \frac{FA}{FA + GR}, \quad \text{FRR} = \frac{FR}{GA + FR}, \quad \text{GAR} = 1 - \text{FRR}.$                        |
| Equal error                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | $\text{FAR}(t^*) = \text{FRR}(t^*).$                                                                                                |
| Open set                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | $\text{FNIR}(t) = 1 - \text{DIR}(t, 1);$ FPIR is the fraction of non-gallery probes whose best gallery score reaches the threshold. |
| Closed set                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | $\text{RR} = \text{CMS@1};$ CMS@ $k$ is the probability that the correct identity is within the first $k$ candidates.               |